SANDEEP NADELLA

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SENIOR SOFTWARE ENGINEER

Senior Software Engineer with 11+ years of experience designing enterprise integration platforms, distributed backend systems, and cloud-native services utilizing various tech stacks involving NodeJS, Java. Experienced in event-driven architectures, APIs, performance optimization, and cross-functional technical leadership, with a growing focus on AI-powered engineering platforms.

SKILLS

Technical Leadership | System Design | Algorithms | Critical Thinking Skills | Mentoring | Cross-Functional Collaboration

TECHNICAL SKILLS

Programming Languages: Java | JavaScript | SQL | Python

Technologies, Libraries, Tools: Node.js | React.js | Prisma ORM | Kubernetes (K8s) | Claude Code | HTML5 | CSS | Bootstrap | Angular (TypeScript) | JSON Web Tokens (JWT) | OAuth | Azure | Spring Boot | Oracle ATG Web Commerce platform | J2EE and Web design | REST, SOAP web services | JSON/XML | Claudia.js | Amazon Web Services (AWS) | AWS Elastic Kubernetes Service(Amazon EKS) | AWS Lambda | AWS Bedrock API Gateway | Bamboo CI/CD | OpenSearch Observability | Android App Development SDK | XSLT | Jasmine testing | JUnit testing | WebRTC | MySQL | PostgreSQL | MariaDB | MongoDB | NoSQL | Python - Django framework | TensorFlow APIs | Git, Perforce code version control tools | Gradle, Maven build tools | OpenCV | Docker | Google Firebase ML Kit | Figma/Axure RP for Prototyping

PROFESSIONAL EXPERIENCE

Align Technology, San Jose, CA **June 2020 - October 2026**

Senior Software Engineer

Led backend architecture and delivery at Align across Consumer Platform, Global Invisalign Experience, Enterprise Engineering Teams, Invisalign Web and Mobile apps, Virtual Care, and Vivera Retainer Subscription, Treatment Planning products. Mentored engineers and guided teams through design and delivery phases.

Leadership and Technical Ownership

- Guided and mentored Developers, QA, Analysts, PMs across experience levels over project releases and ensured product growth, successful timely deliveries. Proposed scalable, fault-tolerant, maintainable designs across services
- Led Align team members and contractors in technical discussions, took ownership of investigations, and developed solutions to identify and fix production issues
- Stabilized, and maintained GraphQL, NodeJS, NestJS using Prisma ORM and Ruby on Rails code repos. Identified architectural risks early in the development lifecycle, proposing design improvements to reduce technical debt and future rework

High Business Impact and Innovation

- Architected, built, owned end-to-end architecture and the delivery roadmap of fully customizable Data Integration Hub to replace TIBCO using node sandbox eliminating the need for point-to-point integrations and specialized integration engineers saving $1Million+ in operating costs
- Built an AI-assisted playground and test-generation framework to simulate integration workflows during development, before deployment to development and staging environments
- Automated integration-workflow code generation with AWS Bedrock models; built a Model Context Protocol (MCP) server enabling local LLMs to construct no-code integrations from user inputs
- Built Data Integrity Check dashboards for identifying and conducting large scale data corrections on production systems through AI assisted coding using Cursor Composer, Claude Code and integrated with Large Language Models(LLMs) in AWS Bedrock for mappers
- Designed Subscription-based payment models for use in Vivera Retainers, Align Diagnostics, and Treatment Planning at Invisalign offered by 100K+ doctors while working with global teams
- Improved performance of API endpoints by over 40% in some cases across services through SQL, Application layer(ActiveRecord), and API design optimizations, and through requirement cut-downs when necessary while ensuring system scalability and built backends to serve metric reports

Technical Architecture and Engineering

- Built an AI incident response system that uses retrieval-augmented generation (RAG) to surface relevant source code blocks and Large Language Models (LLMs) to generate detailed root cause analyses and recommend fixes

- Designed an on-demand log triage workflow to correlate events across services and condense large log volumes into incident timelines that fit within the model's context window
- Developed Salesforce connectors to exchange CDC, platform events over push topics, PubSub API with Confluent Kafka topics
- Integrated Apache Avro schema support into Data Integration Hub connections
- Proposed failed-event reprocessing using sandbox exceptions propagated to stage-level handlers and asynchronous database logging of failures. Reduced failed messages in retry queue by more than 90% incorporating Kafka keys into designs for partition routing, fixing unexpected kubernetes pod restarts, handling authentication failures due to Salesforce/Redis downtimes gracefully
- Designed and built fault-tolerant order management service for processing, fulfilling orders placed in CPQ systems like Bit2Win. Designed Model Context Protocol(MCP) server for accessing read-only metrics
- Designed and built GraphQL schemas, resolvers, used dataloaders, RESTful and Non-RESTful APIs, Kafka publish-subscriber models for various features spanning across Virtual Care, Subscriptions Portal, Align Special Markets Portal, Align Diagnostics Subscription, Align Admin Back Office, My Invisalign Web, iOS, Android applications
- Significantly contributed towards Ruby on Rails, NodeJS microservices at the API, Schema, and System Design levels. Modernized tech stack from RoR to Node
- Supported deployment of docker container stack over Heroku, AWS through Bamboo CICD ensuring Checkmarx, Snyk Security fixes, Unit Test Case integrations. Reported and fixed several security issues in services across Align
- Designed GraphQL resolvers, schemas, backend APIs and assisted in the development of the Aligner fit feature
- Built monitoring dashboard application for analyzing the performance metrics of applications deployed on Heroku. Tech stack includes React, NodeJS, Postgres and Redis

Arizona State University, Tempe, AZ **February 2019 - May 2020**

Full Stack Software Engineer

Drove the design, development and deployment of student grade analytics applications across academic programs, high-performance computing department.

- Designed and built Analytics applications for Student grades for ASU SoMSS, AMLSS departments, MTBI program using Node.js, Angular, Cube.js and deployed in AWS
- Developed core application features including JSON Web Token(JWT) based application login, Secure file storage in AWS S3, Visual analytics dashboard using Cube.js and Chart.js

Oracle Corporation, Hyderabad, India **June 2015 - July 2018**

Senior Software Developer

Designed and delivered enterprise integrations for Oracle Commerce Cloud, Oracle Integration Cloud Services, Order Management and CRM systems working as part of Commerce Cloud Integrations Team. Built backend services and APIs to synchronize orders and customer data across platforms.

- Led Commerce-Responsys: Orders, Profiles and Abandoned Cart sync flows; Commerce-SiebelCRM: Order sync flows; Commerce-Oracle Engagement Cloud for Communications: Order API and Webhook flows; Commerce-ProductHub: File Upload/Download, Products Import/Export flows
- Developed Commerce Cloud Adapter by working on Oracle Integration Cloud Services (ICS) SDK
- Implemented and maintained Swagger functionality of ICS/OIC CC adapter
- Designed XSLT templates for mapping data for Oracle FOM, OROMS, Commerce, Product Hub integration
- Formulated Swagger template based NodeJS custom extensions development for Commerce Cloud
- Led the successful migration of Commerce Cloud API documentation to JSON Hyper-Schema, OpenAPI Specification
- Contributed to Commerce-Responsys, Commerce-SiebelCRM, Oracle Engagement Cloud integrations
- Developed the front end and back end designs for CC-OROMS integration using Java and JavaScript

EDUCATION

MCS, Master of Computer Science

Arizona State University, Tempe; United States GPA: 4/4

B.Tech, Bachelor of Technology

National Institute of Technology, Warangal (NITW), India GPA: 3.6/4